

Q4

Pseudocode:

```

START
  DECLARE quantity : INTEGER
  DECLARE pricePerItem ; subtotal , discount : FLOAT
  DECLARE tax , discountAmount , final BILL : FLOAT.

  INPUT quantity
  INPUT pricePerItem
  INPUT discount
  INPUT tax

  IF quantity <= 0 THEN
    OUTPUT "Error, Quantity must be greater than 0"
  ENDIF

  IF pricePerItem <= 0 THEN
    OUTPUT "Error, the price must be greater than 0"
  ENDIF

  IF discount < 0 OR discount > 100 THEN
    OUTPUT "Error , the discount must b/w 0 and 100"
  ENDIF

  IF tax < 0 OR tax > 100 THEN
    OUTPUT "Error , tax must be b/w 0 and 100"
  ENDIF

  subtotal ← quantity * pricePerItem
  discountAmount ← subtotal - ((subtotal * discount) / 100)
  Final BILL ← discountAmount + ((discountAmount * tax) / 100)

  OUTPUT " _____ Shopping BILL _____ "
  OUTPUT " Quantity : " , quantity

```

OUTPUT "Price per Item: ", pricePerItem

OUTPUT "Subtotal: ", subtotal

OUTPUT "Discount: ", discount

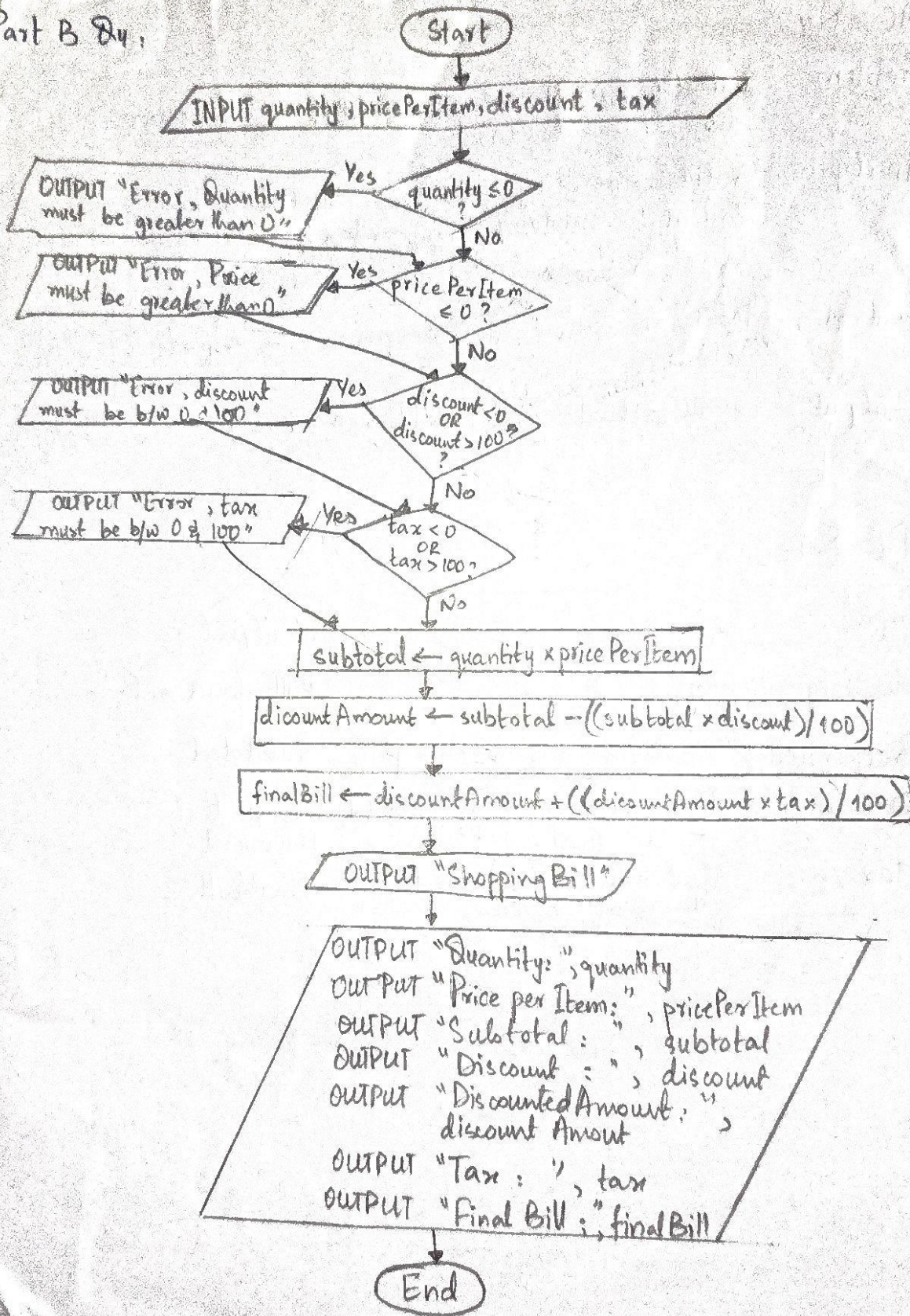
OUTPUT "Discounted Amount: ", discountAmount

OUTPUT "Tax: ", tax

OUTPUT "Final Bill: ", finalBILL

END

Part B Q4:



IPO and PAC chart:

PAC Q4:

Problem	Calculate a customer's final shopping bill after discount and tax.
Analysis	Validate quantity, price, discount and tax. Calculate subtotal, discounted amount and final bill in a sequence.
Control	Decision statements for validating input values.
Output	Bill details and final bill amount.

IPO Q4:

Input	Process	Output
Quantity	Validate values	Bill document
Price per Item	$\text{Subtotal} = \text{quantity} \times \text{price}$	Subtotal
Discount %	$\text{Discounted amount} = \text{subtotal} - \text{discount}$	Discounted Amount
Tax %	$\text{Final bill} = \text{discounted amount} + \text{tax}$	Final bill